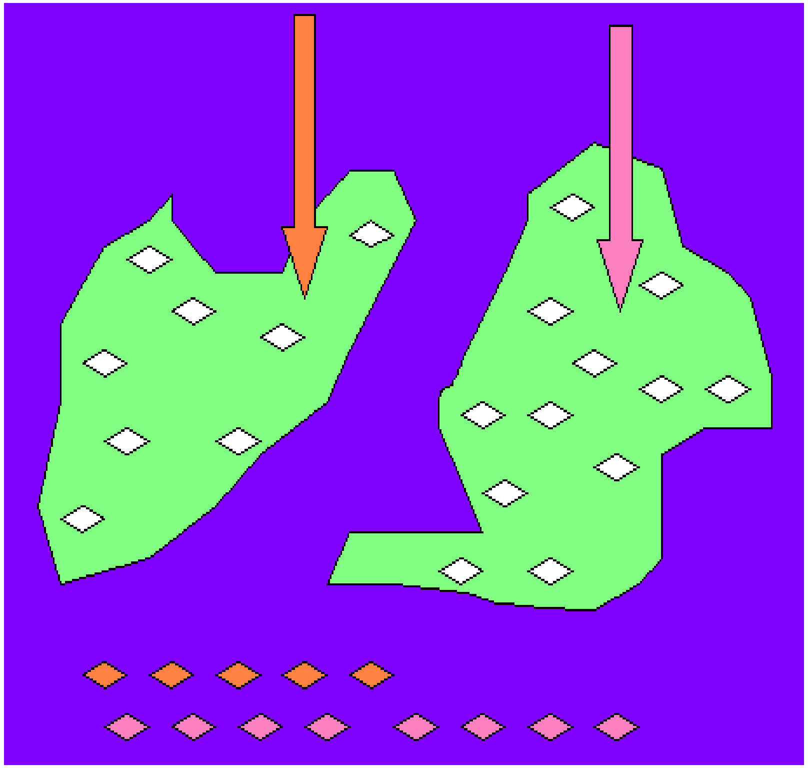
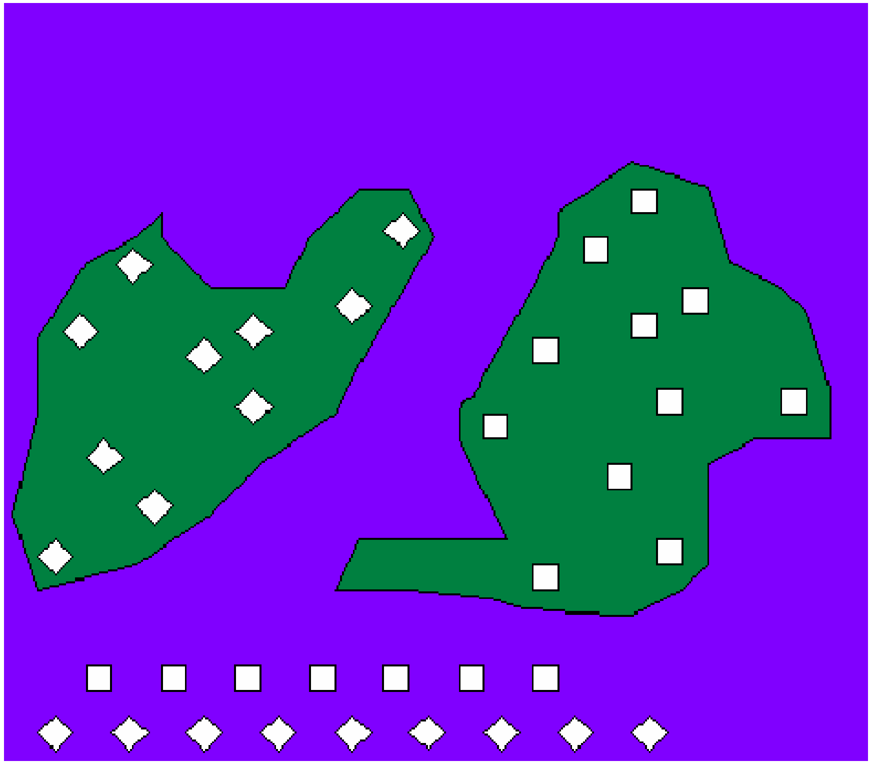
Hypothesis tests for two independent samples

- Compare two proportions
- Compare mean values of two populations

B. Two independent samples model





Model of two groups of objects with different

a) Intervention levels,
b) Individual proper

Compare two proportions – the case of large sample sizes (using Normal distribution)

Let $(X_1, X_2, ..., X_{n_1})$ be a sample of a binary variable X taking value 1 with probability p_1 and value 0 with probability $(1 - p_1)$, $(Y_1, Y_2, ..., Y_{n_2})$ be a sample of a binary variable Y taking value 1 with probability p_2 and value 0 with probability $(1 - p_2)$; $p_1, p_2 \in (0,1)$.

Consider the Hypothesis H: $p_1 = p_2$

and Alternative Hypothesis K: $p_1 \neq p_2$

Note. Variable X has expectation p_1 and variance $p_1(1-p_1)$.

Variable Y has expectation p_2 and variance $p_2(1-p_2)$.

Therefore we can treat the testing problem as a special problem of comparing two mean values (expectations) p_1 and p_2.

By Moivre-Laplace Theorem, for large sample size,

$n_1 \times p_1 \geq 5$ and $n_1 \times (1-p_1) \geq 5$,

$n_2 \times p_2 \geq 5$ and $n_2 \times (1-p_2) \geq 5$,

the sample proportions $m(p_1)/n_1$ and $m(p_2)/n_2$ of appearance of number 1 have distributions approximate to normal distribution with expectation p_1, p_2 and variance $p_1 \times (1-p_1)/n_1$, $p_2 \times (1-p_2)/n_2$, respectively. Denote $m_1 = m(p_1)$ and $m_2 = m(p_2)$.

If the Hypothesis H is true then use the two samples $(X_1, X_2, ..., X_{n_1})$ and $(Y_1, Y_2, ..., Y_{n_2})$ as samples collected from one variable and estimate the common variance of X and Y by

$$\frac{m_1+m_2}{n_1+n_2}.(1-\frac{m_1+m_2}{n_1+n_2}) = \frac{m_1+m_2}{n_1+n_2}.\frac{n_1+n_2-m_1-m_2}{n_1+n_2}$$

then perform a statistic

$$u = \left(\frac{m_1}{n_1} - \frac{m_2}{n_2}\right) / \left[\sqrt{\frac{m_1+m_2}{n_1+n_2}.\frac{n_1+n_2-m_1-m_2}{n_1+n_2}}.\sqrt{\frac{n_1+n_2}{n_1.n_2}}\right]$$

for testing, where m_1 and m_2 respectively are the numbers of values 1 appeared in the above two samples.

By Central Limit Theorem, when sample sizes are large, the difference Mean(X) - Mean(Y) has a distribution very close to Normal distribution. Then the testing procedure can be as follows:

Step 1. Calculate value of statistic

$$u = \left(\frac{m_1}{n_1} - \frac{m_2}{n_2}\right) / \left[\sqrt{\frac{m_1 + m_2}{n_1 + n_2} \cdot \frac{n_1 + n_2 - m_1 - m_2}{n_1 + n_2}} \cdot \sqrt{\frac{n_1 + n_2}{n_1 . n_2}}\right]$$

Step 2. Taking Normal distribution N(0,1) find the probability (p-value)

$$b = \boldsymbol{P}\{\,|\,\boldsymbol{N(0,1)}\,|\ >\ |\,\boldsymbol{u}\,|\,\}$$

Step 3. Compare the probability b (p-value) to the given ahead significance α

* If $b \geq \alpha$ → Accept Hypothesis H, confirm the equality of two proportions

* If $b < \alpha$ → Reject Hypothesis H and conclude two proportions to be different

(One-tail tests can be done similarly)

Version B. Using Normal critical value

Looking in Table of Normal distribution find out critical value $u_{\alpha/2}$ of Normal distribution (the critical value for $\alpha = 5\%$ equals 1.96)

Decide

- Reject Hypothesis **H** if

$$|u| \geq u_{\alpha/2}$$

- Accept Hypothesis **H** if

$$|u| < u_{\alpha/2}$$

Version C. Using confidence intervals

Use confidence intervals (with significance level of α) of estimated proportions for testing:

$$\left[\frac{m_1}{n_1} - Z_{1-\alpha/2} * \sqrt{\frac{m_1}{n_1}(1-\frac{m_1}{n_1})/n_1}\,; \frac{m_1}{n_1} + Z_{1-\alpha/2} * \sqrt{\frac{m_1}{n_1}(1-\frac{m_1}{n_1})/n_1}\right]$$

$$\left[\frac{m_2}{n_2} - Z_{1-\alpha/2} * \sqrt{\frac{m_2}{n_2}(1-\frac{m_2}{n_2})/n_2}\,; \frac{m_2}{n_2} + Z_{1-\alpha/2} * \sqrt{\frac{m_2}{n_2}(1-\frac{m_2}{n_2})/n_2}\right]$$

Decide

Reject Hypothesis **H** **if** the two intervals disjoin

Accept Hypothesis **H** **if** the two intervals have nonempty intersection

Compare two mean values

Let $(X_1, X_2, \ldots, X_n)$ be a sample of n independent observations from a variable X with expectation μ_1 and variance σ^2

$(Y_1, Y_2, \ldots, Y_m)$ be a sample of m independent observations from a variable Y with expectation μ_2 and variance σ^2

Problem: Compare two expectations μ_1 and μ_2 .

→ Estimate and compare two mean values $\bar{X}$ and $\bar{Y}$.

The problem can be solved by using the following Theorem:

Theorem. *Let* $(X_1, X_2, \ldots, X_n)$ *and* $(Y_1, Y_2, \ldots, Y_m)$ *be two samples of n independent observations selected correspondingly from a variable X with sample mean* $\overline{X}$ *and sample variance* S_X^2 *and from a variable Y with sample mean* $\overline{Y}$ *and sample variance* S_Y^2 *(both variables are* ***normal distributed with common variance****). If the hypothesis H is true (*$\mu_1 = \mu_2$*) then the variable (statistic)*

$$t = \sqrt{\frac{n \cdot m}{n+m}} \cdot \sqrt{\frac{n+m-2}{(n-1) \cdot S_X^2 + (m-1) \cdot S_Y^2}} \cdot (\overline{X} - \overline{Y})$$

has Student distribution with (n+m-2) degrees of freedom.

Hypothesis Tests

Hypothesis

$$H: \quad \mu_1 = \mu_2$$

Alternative Hypothesis

$$K: \quad \mu_1 \neq \mu_2$$

Steps of testing

Step 1. Estimate sample mean values $Mean(X)$, $Mean(Y)$ and sample variances $Var(X)$, $Var(Y)$

Step 2. Calculating perform the quantity

$$t = \sqrt{\frac{n.m}{n+m}} \cdot \sqrt{\frac{n+m-2}{(n-1)\cdot Var(X)+(m-1)\cdot Var(Y)}} \cdot (Mean(X)-Mean(Y))$$

Step 3 (p-value approach). Taking a variable $\boldsymbol{T^{(n+m-2)}}$ of Student distribution with (n + m - 2) degrees of freedom calculate the p-value (probability)

$$b = P\ \{\ |\ T^{(n+m-2)}\ |\ \geq\ |\ t\ |\ \}$$

Step 4. Compare the p-value b with a given ahead significance level α (=5%, 1%, 0.5% or 0.1%):

+ If $b \geq \alpha$ → accept Hypothesis **H** and conclude

$$\mu_1 = \mu_2$$

+ If $b < \alpha$ → reject Hypothesis **H** and confirm

$$\mu_1 \neq \mu_2$$

Version B. Using Student critical value

Calculate the critical value $T^{(n+m-2)}_{(1-\alpha/2)}$ of Student distribution with $n+m-2$ degrees of freedom (α is a given ahead significance level =5%, 1% or 0.5%)

Decide

- Reject Hypothesis **H** if

$$|t| \geq T^{(n+m-2)}_{(1-\alpha/2)}$$

- Accept Hypothesis **H** if

$$|t| < T^{(n+m-2)}_{(1-\alpha/2)}$$

Version C. Using confidence intervals

When degree of freedom (sample size) is large, Student distribution approximates Normal distribution. Then we can use confidence intervals (with significance level of 5%) for testing:

$$\left(Mean(X) - T_{(1-\alpha/2)}^{(n-1)} \cdot \frac{SD(X)}{\sqrt{n}}; Mean(X) + T_{(1-\alpha/2)}^{(n-1)} \cdot \frac{SD(X)}{\sqrt{n}}\right),$$

$$\left(Mean(Y) - T_{(1-\alpha/2)}^{(m-1)} \cdot \frac{SD(Y)}{\sqrt{m}}; Mean(Y) + T_{(1-\alpha/2)}^{(m-1)} \cdot \frac{SD(Y)}{\sqrt{m}}\right)$$

Decide

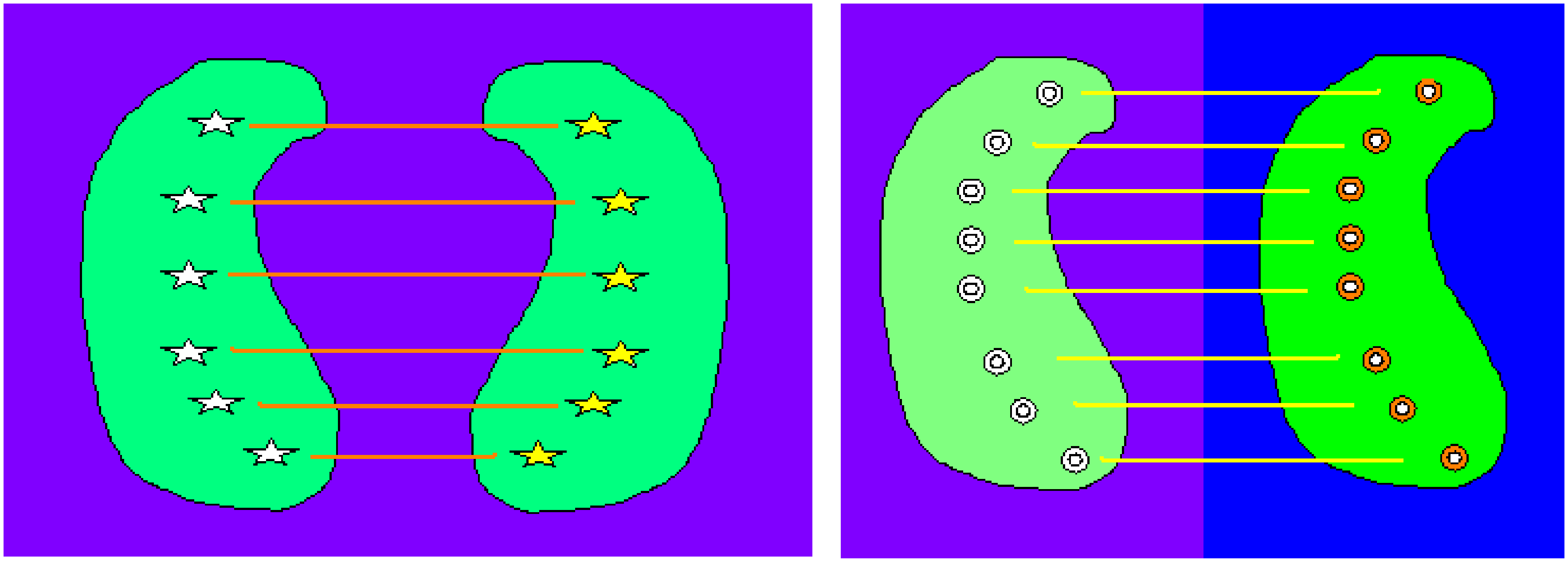
Reject Hypothesis **H** **if** the two intervals disjoin

Accept Hypothesis **H** **if** the two intervals have nonempty intersection

Test for two related (paired) samples

- **Compare two mean values**

C. Model of two dependent (paired) samples



- Two dependent samples model is used in a study when
- A) Each object in the first sample is chosen together with a **similar** (paired) object in the second sample, or
- B) Any object in the second sample is the **same** one in the first sample, but the measures in the two samples are taken under **different conditions**.

Compare mean values of two related samples

For related variables X and Y, the comparison of mean values is equivalent to the comparison the mean value of the difference variable $X - Y$ to value 0 → the problem reduces to ***one-sample model***.

Compare mean values of two related samples

Hypothesis $\quad \mathrm{H}: \mu_1 = \mu_2$

Alternative Hypothesis $\quad \mathrm{K}: \mu_1 \neq \mu_2$

where μ_1 and μ_2 are the expectations of X and Y

With $\xi = X - Y$,

➔comparing expectations μ_ξ of ξ to 0:

Hypothesis $\quad \mathrm{H}: \mu_\xi = 0$

Alternative Hypothesis $\quad K: \mu_\xi \neq 0$

Compare mean values of two related samples

With the empirical value of the test statistic

$$t = \frac{\sqrt{n-1}}{s_{\xi}}\bar{\xi}_n$$

a) Compare the empirical value of the t-test statistic with the critical value $t_{cr} = T_{1-\alpha/2}^{(n-1)}$, which is the $1-\alpha/2$ percentile of the Student distribution with *n*-1 degrees of freedom:

- \- If $|t| \geq t_{cr}$ ➔ reject the hypothesis *H* ,
- \- If $|t| < t_{cr}$ ➔ accept *H* .

Compare mean values of two related samples

b) Taking a random variable $\boldsymbol{T}$ having Student distribution with n-1 degrees of freedom, calculate the probability of significance

$$p_{sig} = \mathrm{P}\{|T| > |t|\}$$

Compare the probability of significance p_{sig} with the significance level α :

- \- If $p_{sig} \leq \alpha$ ➔ reject H,
- \- If $p_{sig} > \alpha$ ➔ accept H.

Compare mean values of two related samples

c) Determine the 95% confidence interval of the estimation $\bar{\xi}_n$:

$$a\%CI(\bar{\xi}_n) = (\bar{\xi}_n - T_{1-\alpha/2}^{(n-1)} \cdot \frac{s_\xi}{\sqrt{n-1}} ; \bar{\xi}_n + T_{1-\alpha/2}^{(n-1)} \cdot \frac{s_\xi}{\sqrt{n-1}})$$

Compare 0 to the confidence interval:

- \- If $0 \notin a\%CI(\bar{\xi}_n)$ ➔ reject *H*,
- \- If $0 \in a\%CI(\bar{\xi}_n)$ ➔ accept *H* .